

**PREDICTING DEPRESSION WITH REDUCED VARIABLES USING
MACHINE LEARNING-BASED METHODS**



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CHAPTER 1

INTRODUCTION

1.1 Background and Rationale

Depression is considered as a common psychiatric disorder. It can occur to a person who is unable to cope with stress, especially in a school age. There are many factors related to it (e.g. Biological, Genetic, Psychological, or environmental). High expectations from parents, too many lessons from school, and entering a different environment are examples of causes of depression in students. Depression may bring aggressive behaviors and may eventually lead to the decision of ending their life. Some students may have just started having symptoms, some students may have severe symptoms. The remedy for this disease is a sensitive consultation in both the way of Psychotherapy and treatment with medicine. Therefore, we created this project to explore and predict which conditions students are in somatization, depression, anxiety, psychotic, or social function.

The data was obtained from the student's depression risk assessment questionnaire. But because the data in the questionnaire has a lot of translators, Therefore, we will use machine learning to help reduce the number of useless or irrelevant variables in the predictions. But the results obtained from the predictions must have a constant accuracy value. In order to predict the correct result, the university will have to find a solution to the next problem. and in order to reduce student risk as much as possible

Machine learning (ML) is the way of creating a computer system to learn by entering data and getting the result out. We do not expect to get the answer from this ML, but we expect to get the output after we input some data through the software which is processed to analyze. There are 3 types of machine learning: supervised learning, unsupervised learning, reinforcement learning.

In this project, there is data obtained from the “Depression Risk Assessment Questionnaire” of case study by Mae Fah Luang University students. and because the number of questions in the mental health assessment questionnaire is too high. Sometimes it can make the survey taker become bored or skip certain questions. So the project we will be able to use machine learning to help create a model of processing of usable or unusable variables to predict the results. The result must have a non-diminished accuracy value in order to predict the correct results, so the university will find a solution and to reduce the risk of students as much as possible.

1.2 Objective

1.2.1 To obtain a predictive model that can predict a mental health status

1.2.2 To see whether unrelated or less useful variables can be reduced in predicting mental condition

1.3 Scope

1.3.1 Case study of Mae Fah Luang University students

1.4 Methodology

1.4.1 Literature review

1.4.2 Data exploration and data analysis

1.4.3 Data cleaning and processing

1.4.4 Initial model building and testing

1.4.5 Design website (Front-end) and database

1.4.6 Developing model and model selection

1.4.7 Model testing

1.4.8 Create website(Back-end)

1.4.9 Combine Model and website

1.5 Plan

Table 1.1 Working plan

[illegible]

Senior project 1 document preparation										
Senior Project 1 presentation										
Developing model & model selection										
Model testing										
Create website (Back-end)										
Combine Model and website										
Senior Project 2 presentation										

1.2 Expected Results

1.2.1 A machine-learning based model that can predict the results with the most accuracy

1.2.2 The knowledge of which questions or which variables are helpful or not for depression assessment

1.3 Place and Equipment

1.2.3 Place

- Mae Fah Luang University
- Dormitory

1.2.4 Software (List all target software and version)

- Anaconda Jupyter Notebook version 6.4.5
- Visual studio version 1.64.2

CHAPTER 2

Literature Review

Major Depressive Disorder (MDD) is one of the most common mental disorders among college students [1]. Depression in students is caused by socioeconomic status[2] and declining academic performance [1]. Significant impairment in role: the risk of dropping out of college rises as a member of a new social group because of the expectations they and their families have created. elevates levels of anxiety [3]. Physical illness reduces physical activity, risky sexual behavior, smoking level, alcohol and drug dependence, quality of life, self-harm behavior, and the risk of suicide. At the university level, some people had anxiety symptoms (45.5%) compared to less educated people (39.2% and 35.8%, respectively) [2]. Predicting the risk of depression Machine learning can be used in processing, where machine learning has many algorithms such as classification, regression, etc.

In[4], sociological characteristics (e.g., age, gender), health (e.g., chronic disease prevalence), quality of life (EuroQol EQ-5D), and logistic regression model normalization using LASSO were used. It is used as a key performance indicator. The result is that LASSO has good classification performance. A model with a few variables performs better than a model with more variables as missing variables is reduced.

The authors of[5], this paper provides 5 different severity level predictions using 8 algorithms (Bayes (NB), Bayes network (BN), k-star, local nearest neighbor (LNN), multilayer. perceptron (MLP), radial basis function network (RBFN), random forest (RF), J48). The results of the statistical measures calculated from the confusion matrix were different. Bayes net performance is better than naïve Bayes because it is a neural network. Therefore, the network performance is better than the general classification.

The author's study [6] involved participants. Females and males of various ages and machine learning used 80% of the depression dataset for training and 20% of the depression dataset for testing. After completing the data set extraction technique, This is due to the presence of unrelated features that decrease performance. So we used three property selection techniques to reduce them: SelectKBest, mRMR, and Boruta. The study used six machine learning classifiers: K-Nearest Neighbor (KNN), AdaBoost, GB, XGBoost, Bagging, and Weighted Voting (e.g., GB, Bagging, and AdaBoost). The results when used with the feature selection technique are as follows: The bag-wrapping classifier showed the best results, with an accuracy of 89.26% without feature selection. Here, KNN showed the least accuracy, with 66.94% accuracy, when comparing the results of different models. It can be concluded that the AdaBoost classifier with the SelectKBest feature selection algorithm is superior to other models. In terms of accuracy, AUC values, ROC lines and other measurable performance metrics, accuracy is up to 92.56%.

This article [7] mentioned that classification is a machine learning algorithm. This predicts the percentage of people suffering from symptoms of stress, anxiety, and depression, according to the level of severity. The dataset was divided into the following ratio: 70:30, representing the training and test sets, respectively. The decision tree part is suitable for predictive problems because it is easy to interpret.

The authors of [8] use information from a depressive survey that already exists. The problem is the limitation of the sample size, which is too small compared to the proper size. Machine learning is therefore used to reduce the variance if the sample size is significantly less than the number of features. As may be the case with many of these studies, nonlinear learning approaches do not have a significant impact on outcomes. And it might be better to use a linear learning method. learning method, so to avoid too much complexity and opportunity to learn, the linear learning method may be optimal. The performance of feature selection is improved with increasing sample size. It may not be appropriate to implement a supervised feature selection method on a small dataset. However, this requires further testing with a larger test set to gain a more concrete understanding of which feature reduction approach is best. This creates an optimal model when given smaller datasets. It is used when available data is limited due to the small sample size, but it is associated with high variance.

CHAPTER 3

ANALYSIS AND DESIGN

3.1 Workflow Overview

Figure X show workflow overview of this project

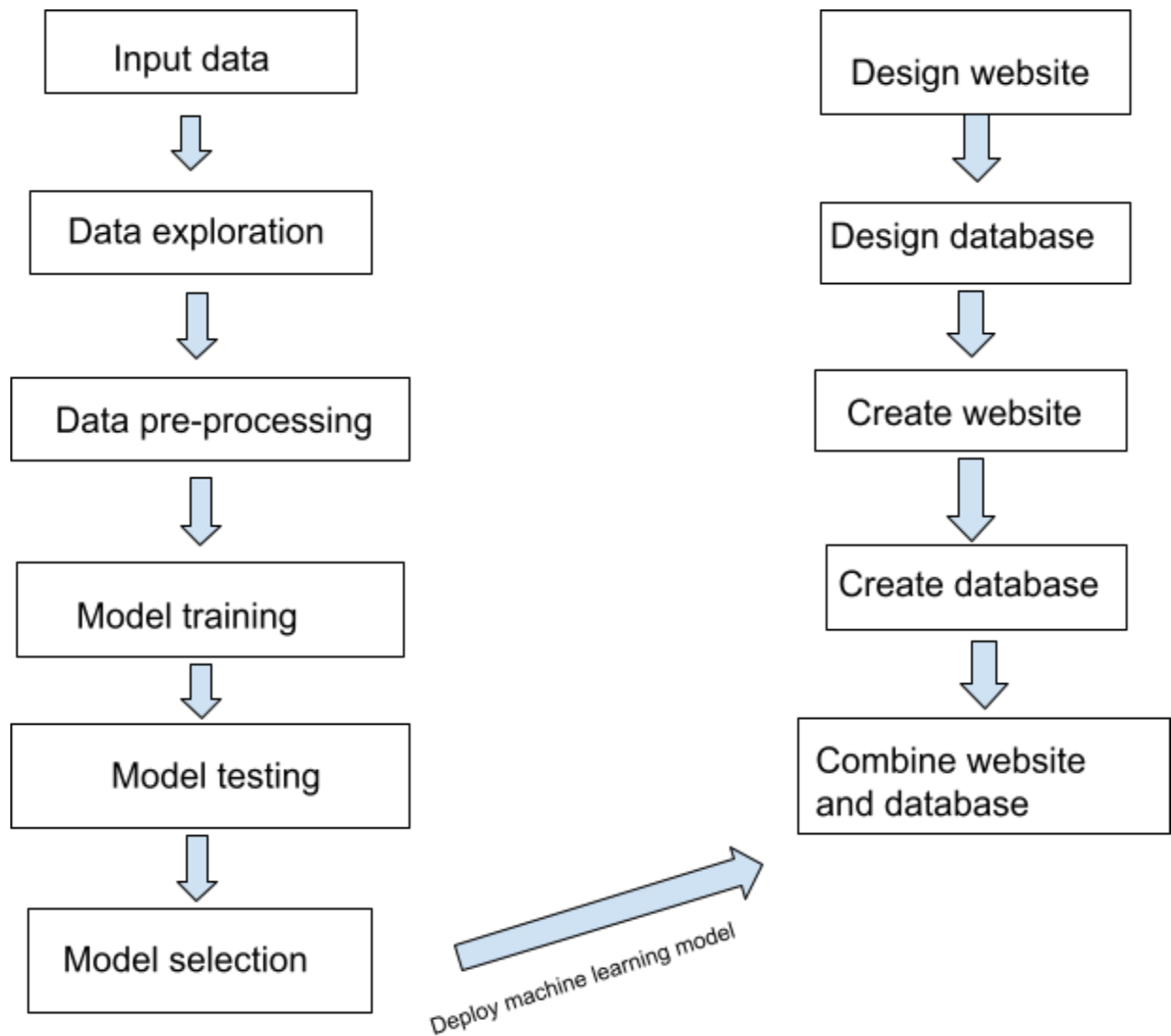


Figure 1.workflow overview

3.1 Data workflow:

Figure 2 shows the process of working in the data section.

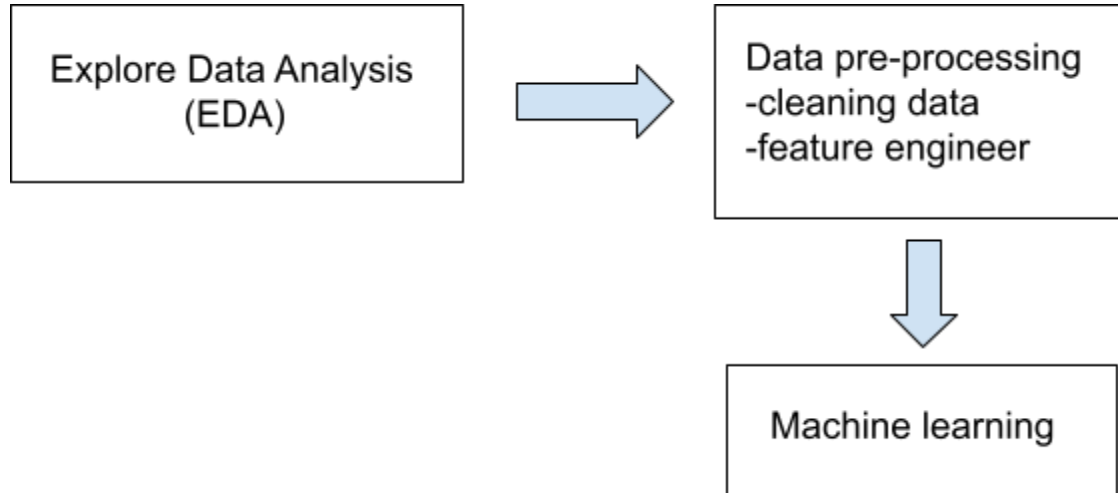


Figure 2.data workflow

3.1.1 Dataset

- This dataset is obtained from a research project questionnaire.

“The Situation and Production System for Undergraduate Graduates of Mae Fah Luang University”, with a total of 4642 people, 811 people with depression and 3831 people without depression. The six part of the information are:

Part1: General Information

Part2: Health measure

Part 3: Emotional Intelligence Assessment

Part4: Learning Style Assessment Form

Part5: Assessment of Cognitive Skills

Part6: Student's 6 Skills Assessment Form according to the Higher Education Qualifications Framework

There are a total of 275 variables. Table 1 shows variable names with explanations

Table 1 variable definition table

order	feature name	description
1	School	School name
2	Major	Major name
3	Year	Year
4	Gender	0=male, 1= female
5	Age	Student age
6	Close_friend	Number of close friends
7	Closefriend_inout	1=in MFU, 2= out of MFU
8	GPAX	GPAX
9	Health_status	physical health
10	Health_status_iden	1=healthy, 2= have symptoms/conditions

11	Family_type	family characteristics
12	Recieve_care	1 father and mother to raise 2 single parents 3 foster relatives 4 Other people who are not relatives to take care of
13	Recieve_care_other	Identify someone other than a parental relative.
14	Family_income	The average family income per month
15	Family_income_enough	1= enough, 2= not enough,3=unknown
16	Student_income	Average student income per month
17	Student_income_enough	1=enough,2=not enough
18	Province	domicile of students
19	Dormitory	1=on campus,2= outside university
20	Relation_fam	relationship with family
21	Relation_friend	relationship with friends

22	Relation_aj	relationship with teachers
23	Feel_uni_1-6	the level of feelings towards the various service departments of the university
24	Feel_envi_7-14	level of feeling towards the environment on campus
25	Relax_activity	frequent relaxation activities
26	Problem	Problems students faced during the last semester 0= does not have, 1=have
27	Problem_study	0= does not have, 1=have
28	Problem_love	0= does not have, 1=have
29	Problem_adjust	0= does not have, 1=have
30	Problem_econ	0= does not have, 1=have
31	Problem_other	0= does not have, 1=have
32	Attitude	Attitudes towards the field of study/profession being studied
33	Mental_1-70	mental health test

34	EQ1-EQ52	Emotional Intelligence Assessment
35	Learning_style_1-30	questionnaire about learning model
36	Reading_1 -Career_37	21st Century Learning Skills Assessment
37	Course_name	Course name
38	Semester	Semester
39	Academic_year	Academic year
40	Field_train_name	internship place name
41	Field_train_duration	Length of time to practice professional experience
42	TQF1.1-6.1	Information on learning assessments according to the National Qualifications Framework for Higher Education

Table 1.

3.1.2 Data subsetting

In the work, we divide the data into 7 sets to check which sets, when predicted, would provide the highest accuracy. Each data set has a different number of variables as follows :

	General Information	Health measure		Emotional Intelligence Assessment	Learning Style Assessm ent Form	Assessment of Cognitive Skills	Student's 6 Skills Assessme nt Form
		All questions	The only question about Depression				
Dataset 1	/	/		/	/	/	/
Dataset 2	/		/	/	/	/	/
Dataset 3	/			/	/	/	/
Dataset 4	/			/	/	/	

Dataset 5	/			/	/		
Dataset 6	/			/			
Dataset 7	/						

Table 2.

3.1.3 Data exploratory-Explore data by making visualizations.

Figure 3-9 shows data graphs for each feature.

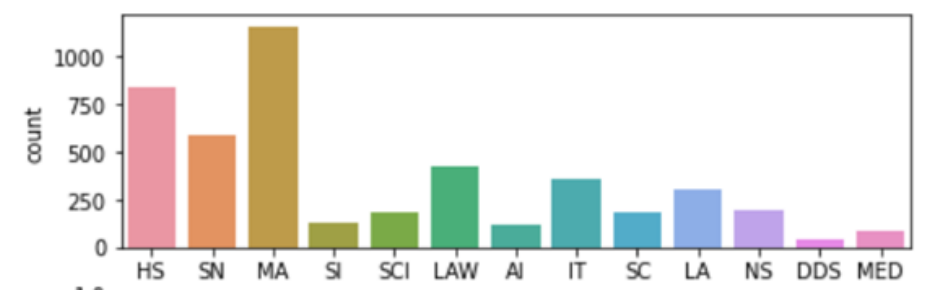


Figure 3. School

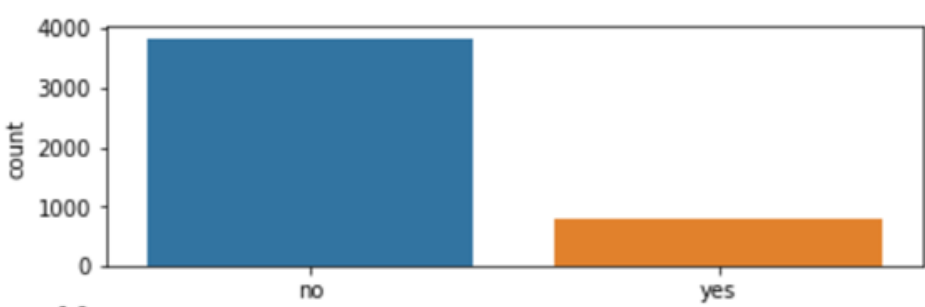


Figure 4. Depression

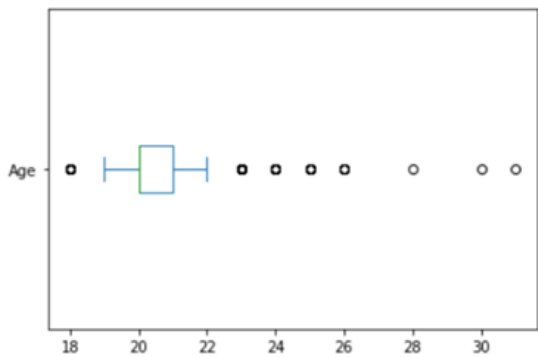


Figure 4. Age

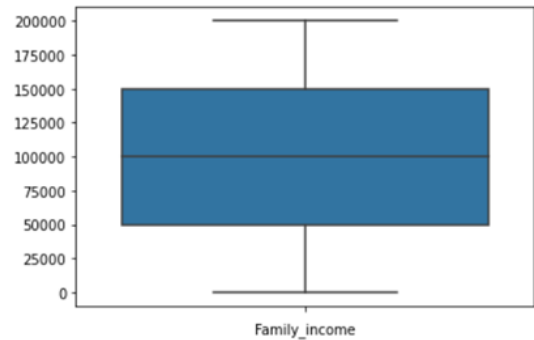


Figure 5. Family_income

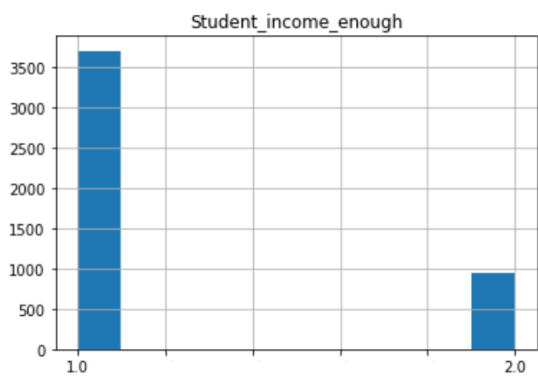


Figure 6. Student_income_enough

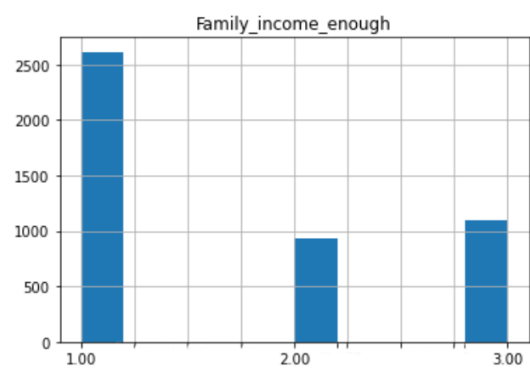


Figure 7. Family_income_enough

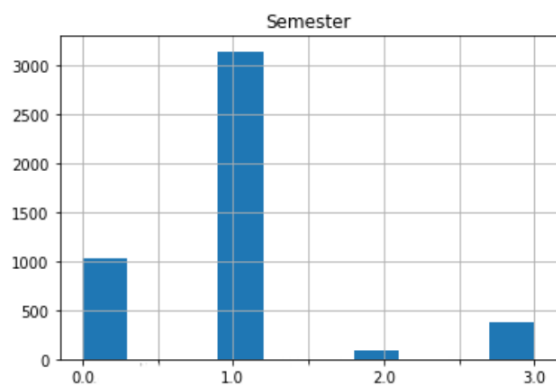


Figure 8. Semester

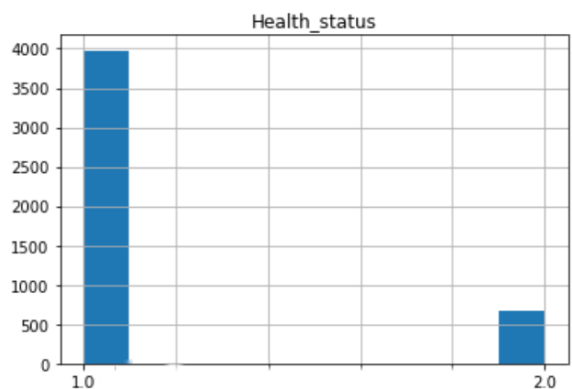


Figure 9. Health_status

3.2 Data pre-processing - Prepare the data before training the model.

Table 3. Show the data stored in numerical.

	Year	Gender	Age	Closefriend_inout	GPAX	Health_status	Family_type	Recieve_care	Family_income
0	2	1	19	1	999.0	1	2	1	248123.643489
1	2	1	20	2	999.0	2	2	1	248123.643489
2	2	1	19	1	2.0	1	2	1	15000.000000
3	2	1	19	1	1.0	1	2	1	248123.643489
4	2	0	19	1	1.1	1	2	2	10000.000000

Table 4. Show the data stored in categorical.

	School	Close_friend	Health_status_iden	Recieve_care_other	Academic_year
0	HS	4	ไม่มี	ไม่มี	-
1	HS	5	ถูกน้ำในรังไข	ไม่มี	2016
2	SN	1	ไม่มี	ไม่มี	2016
3	MA	3	ไม่มี	ไม่มี	2016
4	HS	3-4	ไม่มี	ไม่มี	2016

Table 5. show the data converted from categorical to numerical.

	School_AI	School_DDS	School_HS	School_IT	School_LA	School_LAW	School_MA	School_MED	School_NS
0	0	0	1	0	0	0	0	0	0
1	0	0	1	0	0	0	0	0	0
2	0	0	0	0	0	0	0	0	0
3	0	0	0	0	0	0	1	0	0
4	0	0	1	0	0	0	0	0	0
...	...	...	...	...	...	...	...	...	...
4637	0	0	1	0	0	0	0	0	0
4638	0	0	1	0	0	0	0	0	0
4639	0	0	1	0	0	0	0	0	0
4640	0	0	1	0	0	0	0	0	0
4641	0	0	1	0	0	0	0	0	0

-split data for training and testing

x_train shape: (3249, 594)

y_train shape: (3249,)

x_test shape: (1393, 594)

y_test shape: (1393,)

3.2.1 Variable importance – explain the results

Figure 10 show importance values of each feature

```
Close_friend_28          0.000000
Health_status_iden_ภูมิแพ้ แพ้อากาศ    0.000000
Health_status_iden_ภูมิแพ้ แพ้หน้า      0.000000
Health_status_iden_ภูมิแพ้ เจ็บหัวเข่าจากอุบัติเหตุ 0.000000
Health_status_iden_ภูมิแพ้ หอบหืด      0.000000
...
Mental_17                0.041391
Mental_21                0.043000
Mental_27                0.044912
Mental_20                0.051917
Mental_26                0.065465
Length: 594, dtype: float64
```

Figure 10. importance values

3.3 Web page design and database workflow:

Figure 11 Show web page design and database

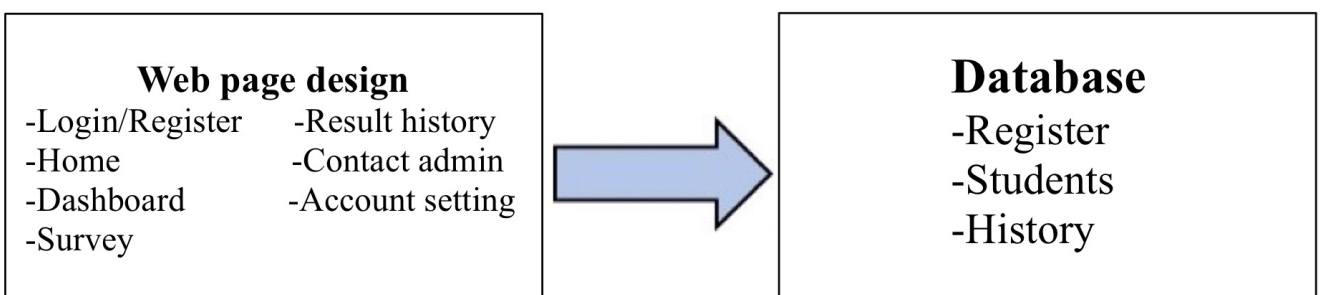


Figure 11. Web and database workflow

3.3.1 Web page view before-after login of user and admin

Figure 12. Shows which pages the admin and user can view.

		Login/Register	Home	Dashboard	Survey	Result history	Contact admin	Account setting
User	before login	✓	✓	✓			✓	
	after login	✓	✓	✓	✓	✓	✓	✓
Admin	view	✓	✓	✓	✓	✓	✓	✓

Figure 12. Bus matrix

3.3.2 Web page design

1.Login/Register

Figure 13. Show page Login/Register for users.

The screenshot displays a web application for Mae Fah Luang University. The browser address bar shows 'localhost:3000/login'. The page layout includes a top navigation bar with the university logo and a 'Log In / Register' button. A left sidebar provides navigation to 'Home', 'Dashboard', and 'Contact Admin'. The main area is divided into two panels: 'Login' and 'Register'. The 'Login' panel contains input fields for 'UserID' and 'Passwords', along with links for 'Forgot password' and 'Contact Admin', and a green 'Login' button. The 'Register' panel includes input fields for 'Name', 'UserID', 'Email', 'Passwords', and 'Confirm Passwords', radio buttons for 'Gender' (Male/Female), and a red 'Register Now' button. A yellow informational box in the Register section specifies: 'User 8 or more characters with a mix of letters,number'. The footer of the page reads '2022-MAE FAH LUANG UNIVERSITY'.

Figure 13. login/Register page

-Register using the information of students in Mae Fah Luang University as follows:
name, student ID, Lamduan Mail, Gender

-Set a password that must be 8 or more characters with a mix of letters and numbers.

-If you click "forgot password", the system will send a password reset link to Lamduan mail.

-Click "contact admin." If there is a problem in filling in the information or others, the admin will fix the problem.

-When logging in, if you enter the wrong user ID and password, the system will alert you that the user ID or password is incorrect.

2.Home

Figure 13 Show home page or invite page.

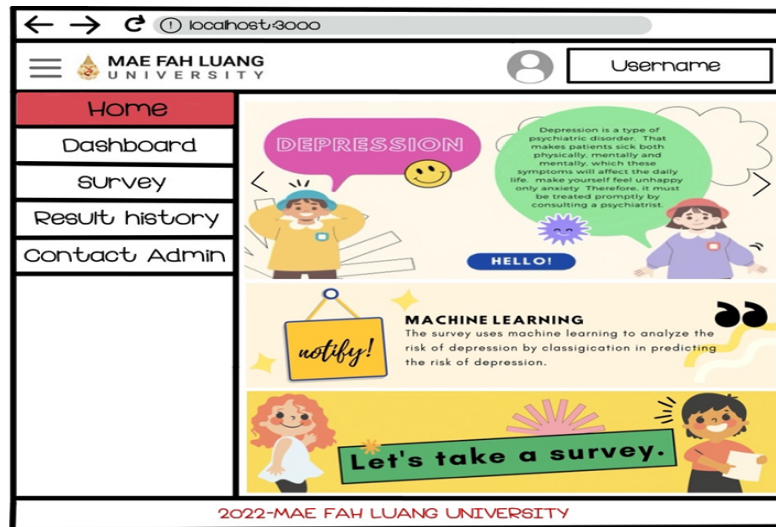


Figure 13. Home page

-It's an invitation to take a survey.

-Describe the details of how machine learning works on this website.

3.Dashboard

Figure 14 Show statistics of Mae Fah Luang University students who have a risk of depression.

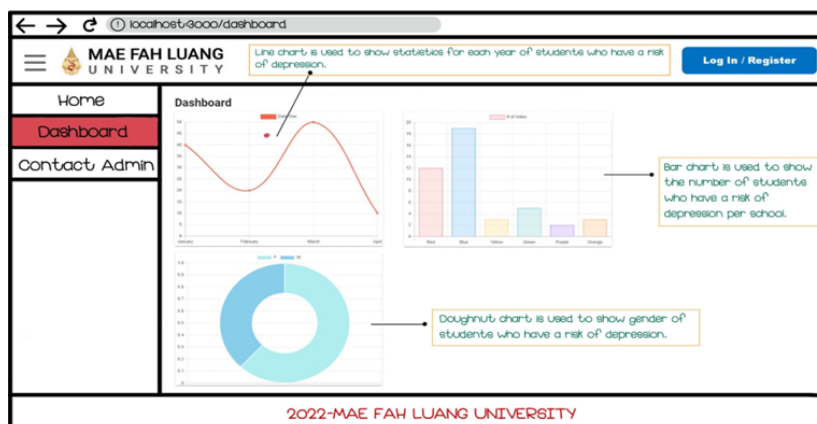


Figure 14. Dashboard page

4. Survey and result

Figure 15. Show the survey page and the survey result page.

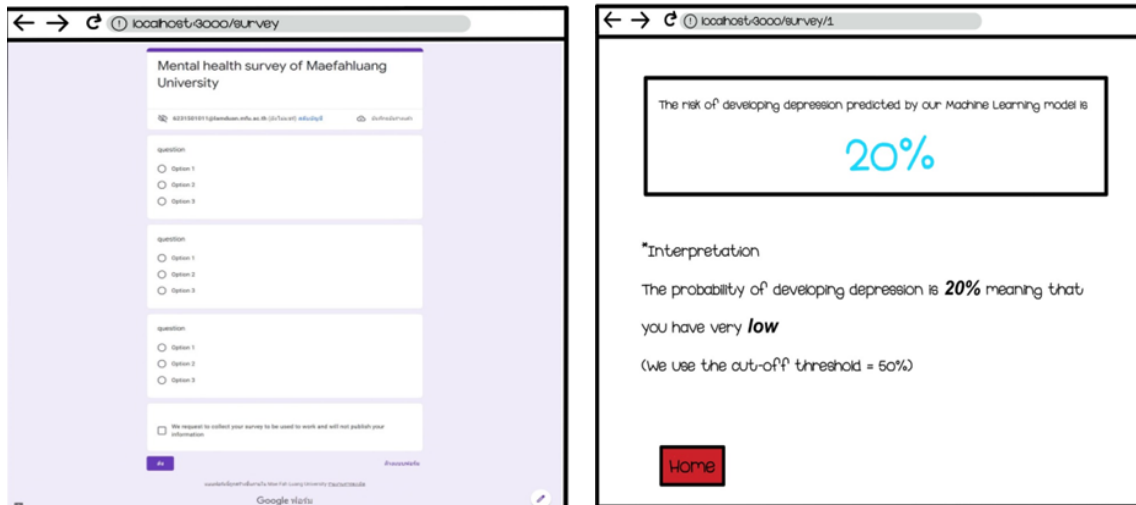


Figure 15. survey page

- Collect data from surveys to predict depression risk with a machine learning model.
- Using the cut-off threshold = 50%

5. Result history

Figure 16. show user results history survey

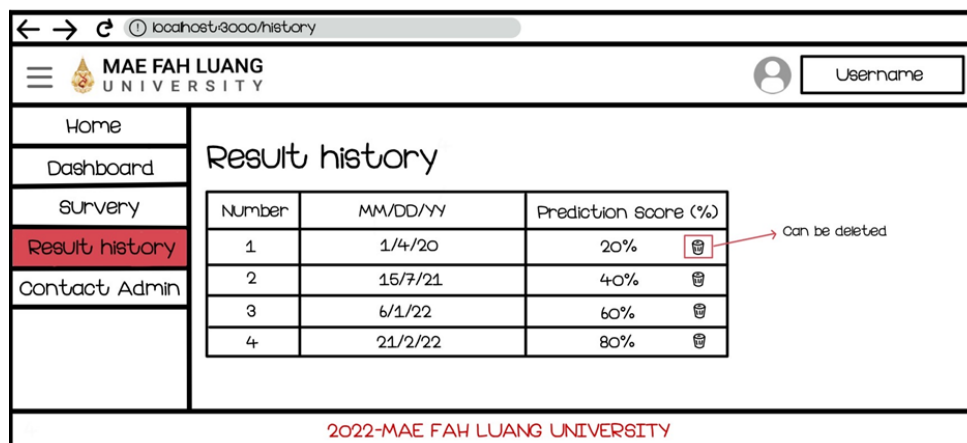


Figure 16. Result history page

- Can delete history.

6.Contact admin

Figure 17 Show contact admin page.

The screenshot shows a web browser at the URL `localhost:3000/contactadmin`. The page header includes the university logo and name, a user profile icon, and a 'Username' input field. A sidebar on the left contains navigation links: Home, Dashboard, Survey, Result history, and Contact Admin (which is highlighted in red). The main content area has a blue background with the text 'CONTACT ADMIN' in large, bold, black letters. To the right of this text are three input fields labeled 'Full Name', 'Email', and 'Message'. Below these fields is a red button labeled 'Contact Us' with the text 'Click Here' underneath it. At the bottom of the main content area, there is a yellow section with the text 'Contact' and the email address 'hi@lamduan.mfu.ac.th', accompanied by Facebook and Messenger icons. The footer of the page displays '2022-MAE FAH LUANG UNIVERSITY'.

Figure 17. Contact admin page

-It is the duty to ask questions, criticize, and propose solutions to problems.

-Use information from name, email, and message of various problems to send information to the admin to fix.

7.Account setting

Figure 18. Show account settings page.

The screenshot shows a web browser at the URL `localhost:3000/1`. The page header is identical to the previous figure, showing the university logo, name, user profile icon, and 'Username' input field. The sidebar on the left contains the same navigation links: Home, Dashboard, Survey, Result history, and Contact Admin. The main content area has a white background with the text 'Account settings' in a large, black, sans-serif font. Below this text are two gray buttons: 'Reset password' and 'Delete account'.

Figure 18. Account setting

-There will be a reset password and a delete account.

3.3.3 Accessibility of admin and user

Admin

- Can have access to everything except the user's password.
- Responsible for reviewing, updating and fixing problems with information within the website.
- Manage access rights for users.
- Respond to user complaints promptly and appropriately.

User

- The user must be a student at Mae Fah Luang University only.
- Must log in to fully access surveys and visit the website.
- Can delete user accounts when they do not want to use that account.
- Can delete the results of the survey history.
- Can change the password.
- User has the right to contact and query Admin if found problems.

3.3.4 Database

Figure 19. Show ER-Diagram of database table as can see below:

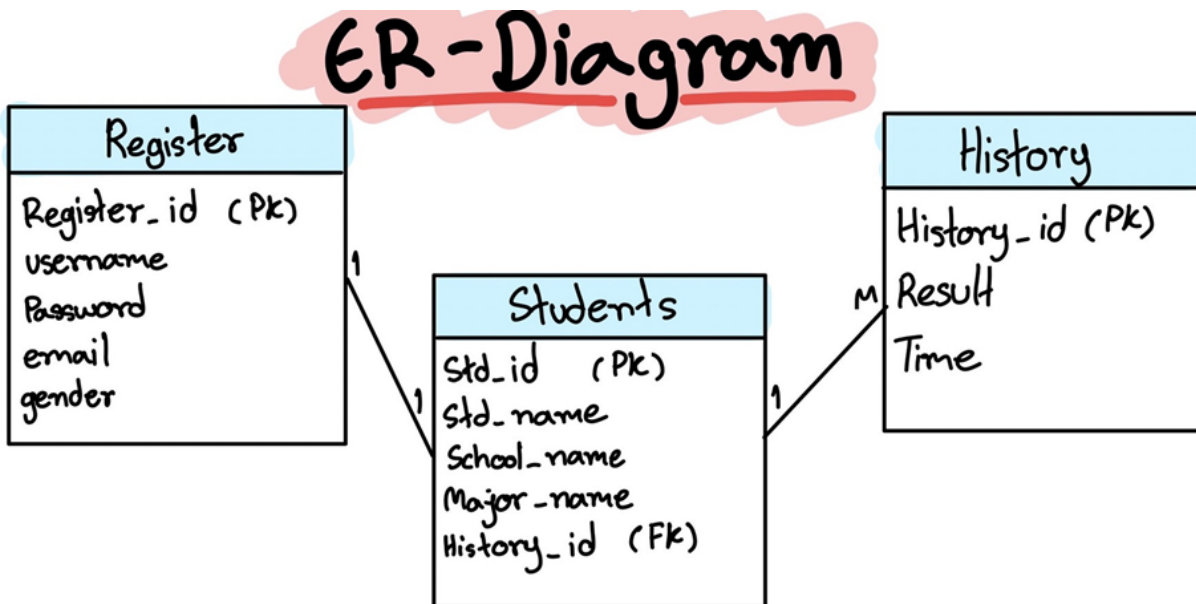


Figure 19.ER-Diagram

Table 3. Show Students_Detail of database table as can see below:

Students - Detail

Attribute Name	Description	Data Type	Key (PK/FK)	Reference Attribute/ Table	Null	Sample Data
Std_id	student ID	INT	PK			
Std_name	student Name	varchar (50)				
School_name	School name	varchar (50)				
Major_name	Major name	varchar (50)				
History_id	Result History	INT	FK	History_id/ History		

Table 3. Student detail

Table 4. Show Register_Detail of database tree table as can see below:

Register - Detail

Attribute Name	Description	Data Type	Key (PK/FK)	Reference Attribute/ Table	Null	Sample Data
Register_id	Register ID	INT	PK			
Username	Student username	varchar (45)				
Password	Student Password	varchar (10)				
email	Landuan mail	varchar (45)				
gender	gender	varchar (10)				

Table 4. Register detail

Table 5. Show History_Detail of database tree table as can see below:

History - Detail

Attribute Name	Description	Data Type	Key (PK/FK)	Reference Attribute/ Table	Null	Sample Data
History_id	History ID	INT	PK			
Result	Result score	varchar (20)				
Time	Time	datetime				

Table 5. History detail

CHAPTER 4

Conclusion

Initial model results :

Take data from the research project questionnaire."The Situation and Production System for Undergraduate Graduates of Mae Fah Luang University" to do an exploration data analysis and create graphs to view statistics of each variable. then do data preprocessing and feature engineering with a one-hot method to transform data. Bring data to share for training and testing models. and try to make predictions

The website is made for the students to be able to come and evaluate themselves. The web page that is designed has a total of 7 pages. Including the design of user and admin usage on the website. The part of the database will design a table to collect student data, usage history and website registration information to use the information on the page Login/Register Result history and Dashboard. And designing website pages to support model machine learning to be deployed on the website

Discussion

From the data from the questionnaire divided into 7 data sets and try to train and test the model. But still get quite a bit of accuracy, so in the next term will develop a model. In order to get more predictive accuracy of the model. As can see result of each dataset below

Figure 20-26 shows accuracy results for each dataset.

Dataset 1 (use all data and all variables), the result of accuracy is

	precision	recall	f1-score	support
no	0.96	0.98	0.97	1150
yes	0.90	0.81	0.85	243
accuracy			0.95	1393
macro avg	0.93	0.90	0.91	1393
weighted avg	0.95	0.95	0.95	1393

Figure 20. Dataset1

Dataset 2 (part of health measure is use only question about depression)
the result of accuracy is

	precision	recall	f1-score	support
no	0.96	0.98	0.97	1150
yes	0.90	0.82	0.86	243
accuracy			0.95	1393
macro avg	0.93	0.90	0.91	1393
weighted avg	0.95	0.95	0.95	1393

Figure 21.Dataset 2

Dataset 3 (use data from General Information, Emotional Intelligence Assessment, Learning Style Assessment Form, Assessment of Cognitive Skills, Student's 6 Skills Assessment Form), the result of accuracy is

	precision	recall	f1-score	support
no	0.88	0.89	0.89	1150
yes	0.45	0.43	0.44	243
accuracy			0.81	1393
macro avg	0.67	0.66	0.66	1393
weighted avg	0.81	0.81	0.81	1393

Figure 22.Dataset 3

Dataset 4 (use data from General Information, Emotional Intelligence Assessment, Learning Style Assessment Form, Assessment of Cognitive Skills),
the result of accuracy is

	precision	recall	f1-score	support
no	0.88	0.89	0.88	1150
yes	0.43	0.40	0.41	243
accuracy			0.80	1393
macro avg	0.65	0.64	0.65	1393
weighted avg	0.80	0.80	0.80	1393

Figure 23.Dataset 4

Dataset 5 (use data from General Information, Emotional Intelligence Assessment, Learning Style Assessment Form), the result of accuracy is

	precision	recall	f1-score	support
no	0.88	0.88	0.88	1150
yes	0.44	0.44	0.44	243
accuracy			0.80	1393
macro avg	0.66	0.66	0.66	1393
weighted avg	0.80	0.80	0.80	1393

Figure 24.Dataset 5

Dataset 6 (use data from General Information, Emotional Intelligence Assessment), the result of accuracy is

	precision	recall	f1-score	support
no	0.88	0.87	0.88	1150
yes	0.42	0.46	0.44	243
accuracy			0.80	1393
macro avg	0.65	0.66	0.66	1393
weighted avg	0.80	0.80	0.80	1393

Figure 25.Dataset 6

Dataset 7 (use data from General Information only), the result of accuracy is

	precision	recall	f1-score	support
no	0.83	0.82	0.82	1150
yes	0.19	0.19	0.19	243
accuracy			0.71	1393
macro avg	0.51	0.51	0.51	1393
weighted avg	0.72	0.71	0.71	1393

Figure 27.Dataset 7

For a website. It's just the design of the website. Next semester to make a real website by combining database + back end + front end together

Reference

[1]Depression and Anxiety During the COVID-19 Pandemic in an Urban, Low-Income Public University Sample (Sasha Rudenstine, Kat McNeal, Talia Schulder, Catherine K. Ettman, Michelle Hernandez, Kseniia Gvozdieva, Sandro Galea, 2021, International Society for Traumatic Stress Studies)

<https://onlinelibrary.wiley.com/doi/10.1002/jts.22600>

[2] Prediction of major depressive disorder onset in college students (David D. Ebert, Claudia Buntrock, Philippe Mortier, Randy Auerbach, Kiona K. Weisel, Ronald C. Kessler, Pim Cuijpers, Jennifer G. Green, Glenn Kiekens, Matthew K. Nock, Koen Demyttenaere, Ronny Bruffaerts,2019, Pubmed)

<https://pubmed.ncbi.nlm.nih.gov/30521136/>

[3]Depression and anxiety among university students in Hong Kong (Kevin WC Lun, CK Chan, Patricia KY Ip, Samantha YK Ma, WW Tsai, CS Wong, Christie HT Wong, TW Wong, D Yan,2018,Hong Kong Academy of Medicine Press)

<https://www.hkmj.org/system/files/hkmj176915.pdf>

[4]Predicting Depression in Community Dwellers Using a Machine Learning Algorithm (Seo-Eun Cho, Zong Woo Geem, Kyoung-Sae, 2021, Diagnostics)

<https://www.mdpi.com/2075-4418/11/8/1429>

[5]Assessment of Anxiety, Depression and Stress using Machine Learning Models (Prince Kumara, Shruti Garga, Ashwani Gargb, 2020, ELSEVIER)

<https://www.sciencedirect.com/science/article/pii/S1877050920311984>

[6]An in-depth analysis of machine learning approaches to predict depression (Md. Sabab Zulfiker , Nasrin Kabir, Al Amin Biswas a , Tahmina Nazneenc , Mohammad Shorif Uddinb, 2021, ELSEVIER)

<https://www.sciencedirect.com/science/article/pii/S2666518221000310>

[7]Predicting Anxiety, Depression and Stress in Modern Life using Machine Learning Algorithms (Anu Priya, Shruti Garg, Neha Prerna Tigga, 2020, ELSEVIER)

<https://www.sciencedirect.com/science/article/pii/S1877050920309091>

[8] Studying depression using imaging and machine learning methods (Meenal J. Patel, Alexander Khalaf, Howard J. Aizenstein, 2016, ELSEVIER)

<https://www.sciencedirect.com/science/article/pii/S2213158215300206>

